

2027 SEAK Pink Salmon Forecast

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Objective

To forecast the Southeast Alaska (SEAK) pink salmon commercial harvest in 2027. This document is for guidance as to what was done for the current forecast year. It is for internal use only.

Executive Summary

Forecasts were developed using an approach originally described in Wertheimer et al. (2006), and modified in Orsi et al. (2016) and Murphy et al. (2019), but assuming a log-normal error structure (Miller et al. 2022). This approach is based on a multiple linear regression model with either 1) the raw juvenile pink salmon CPUE (`adj_raw_pink_log`) or 2) the vessel-adjusted juvenile pink salmon CPUE (i.e., adjusted by the pooled-species vessel calibration coefficient for the Cobb; vessel-adjusted CPUE) as a proxy for abundance. Additional variables included a vessel factor to account for the survey vessels through time, an odd and even year factor to account for potential odd and even year cycles of abundance, and temperature data from either the Southeast Alaska Coastal Monitoring Survey (SECM; Piston et al. 2021; `ISTI20_JJ`) or from satellite-derived sea surface temperature (SST) data (Huang et al. 2017).

There were 54 individual models considered:

- `adj_raw_pink_log`, an odd/even year factor, and a vessel factor (`m1`);
- `adj_raw_pink_log`, an odd/even year factor, a vessel factor, and temperature data from the SECM survey (`m2`);
- `adj_raw_pink_log`, an odd/even year factor, a vessel factor, and satellite-derived SST data (`m3-m18`);
- `adj_raw_pink_log` with a vessel interaction, an odd/even year factor, and a vessel factor (`m1i`);

- `adj_raw_pink_log` with a vessel interaction, an odd/even year factor, a vessel factor, and temperature data from the SECM survey (m2i);
- `adj_raw_pink_log` with a vessel interaction, an odd/even year factor, a vessel factor, and satellite-derived SST data (m3i-m18i);
- vessel-adjusted CPUE and an odd/even year factor (m1a);
- vessel-adjusted CPUE, an odd/even year factor, and temperature data from the SECM survey (m2a); and
- vessel-adjusted CPUE, an odd/even year factor, and satellite-derived SST data (m3a-m18a).

The `adj_raw_pink_log` variable is defined as the natural log transformed juvenile pink salmon catches ($\ln(\text{CPUE} + 1)$) by haul, standardized to a 20-minute trawl set. The CPUE values were averaged over all 8 stations by month; data from whichever month (June or July) had the highest log transformed CPUE is then used as the juvenile abundance index for that particular year. This CPUE variable is not vessel-calibrated. The Steller and Chellissa vessels were only used for one year each, 2008 and 2009, respectively, and so those two years are not used in the models that incorporated the `adj_raw_pink_log` variable (Table 1).

The vessel-adjusted CPUE variable is defined as the natural log transformed juvenile pink salmon catches ($\ln(\text{CPUE} + 1)$) by haul, standardized to a 20-minute trawl set. The CPUE was then multiplied by a pooled-species vessel calibration coefficient, based on calibration of the fishing power of various vessels used in the SECM project to the fishing power of the NOAA ship John N. Cobb, the vessel used for the initial 11 years of the project. The CPUE values were averaged over all 8 stations by month; data from whichever month (June or July) had the highest log transformed CPUE is then used as the juvenile abundance index for that particular year (Table 1). As there was not a SECM survey in May of 2025 or 2026, temperature data from May were unavailable. Therefore, the temperature variable from the SECM survey (`ISTI20_JJ`) was the average from June and July. In prior forecasting years, the ISTI variable was the average from May through July.

The model performance metric one-step-ahead mean absolute percentage error (MAPE) for the last five years (MAPE5; juvenile years 2021 through 2025, harvest years 2022 through 2026) was used to evaluate the forecast accuracy of the 54 individual models. Leave-one-out cross-validation MAPE (LOOCV) was used as a second measure of predictive performance, AICc values were calculated for each model to guard against over-parameterization, and the adjusted R-squared values, significant terms, and overall p -value of each model were used to evaluate fit.

Based upon the 5-year MAPE, the AICc values, significant parameters in the models, and the adjusted R-squared values, model m13a (vessel-adjusted CPUE, an odd/even year factor, and `NSEAK_SST_AMJ`; Appendix C) was selected. The 2027 forecast using this model has a point estimate of 66.7 million fish (80% prediction interval: 41.5 to 97.5 million fish).

Analysis

Individual, Multiple Linear Regression Models

Biophysical variables based on data from Southeast Alaska (Table 1) were used to forecast the harvest of adult pink salmon in Southeast Alaska, one year in advance, using individual, multiple linear regression models with two main structures (structure 1: models m1-m18 and m1i-m18i; structure 2: models m1a-m18a). The response variable was the natural log of the Southeast Alaska adult pink salmon harvest, $Y_t = \ln(C_t)$, where C_t is the harvest in year t (millions of fish). The juvenile abundance indices (adj_raw_pink_log and vessel-adjusted CPUE) are already on a log scale, $\ln(\text{CPUE} + 1)$, and were not transformed further; temperature data were not transformed. The forecast $\hat{Y}_i$ and 80% prediction intervals (based on output from program R; R Core Team 2025) from the 54 regression models were exponentiated, and the point forecast was bias-corrected (Miller 1984),

$$\hat{F}_i = \exp \left(\hat{Y}_i + \frac{\hat{\sigma}_i^2}{2} \right), \quad (1)$$

where $\hat{F}_i$ is the preseason forecast for model i in millions of fish and $\hat{\sigma}_i$ is the residual standard error of model i (i.e., $\hat{\sigma}_i^2$ is the residual variance). The prediction interval bounds were exponentiated without the bias correction.

In the equations below, $V_{1,t}$ and $V_{2,t}$ are indicator variables for the Medea and the NW Explorer (the reference vessel is the Cobb), O_t is an indicator for an odd return year (the reference is an even return year), CPUE_t is the juvenile abundance index, and $\epsilon_t \sim N(0, \sigma^2)$. Models with a temperature variable add a term $\beta_T T_t$, where T_t is the temperature index.

Model structure 1 with a vessel interaction (m1i-m18i)

The first model, m1i, consisted of the juvenile abundance index adj_raw_pink_log with a vessel interaction, an odd/even year factor, and a vessel factor,

$$Y_t = \beta_0 + \beta_1 V_{1,t} + \beta_2 V_{2,t} + \beta_3 O_t + \beta_4 \text{CPUE}_t + \beta_5 (V_{1,t} \times \text{CPUE}_t) + \beta_6 (V_{2,t} \times \text{CPUE}_t) + \epsilon_t. \quad (2)$$

The vessel factor and odd/even year factor adjust the intercept, and the vessel interaction terms adjust the slope; vessel-specific slopes for the juvenile abundance index plus additive effects of vessel and the odd/even year factor. For example, during even return years when the research vessel is the Cobb, the model simplifies to $Y_t = \beta_0 + \beta_4 \text{CPUE}_t + \epsilon_t$. Although the simplest model does not contain a temperature variable, including temperature data with CPUE has been shown to result in a substantial improvement in the accuracy of model predictions

(Murphy et al. 2019). The temperature index for models m2i-m18i was either the SECM survey Icy Strait temperature index (ISTI20_JJ; Murphy et al. 2019) or one of the 16 satellite-derived SST variables (Huang et al. 2017; definitions in Appendix A).

Model structure 1 without a vessel interaction (m1-m18)

Models m1 through m18 were a simplified version of models m1i through m18i without the vessel interaction terms,

$$Y_t = \beta_0 + \beta_1 V_{1,t} + \beta_2 V_{2,t} + \beta_3 O_t + \beta_4 \text{CPUE}_t + \epsilon_t. \quad (3)$$

Models m1 through m18 estimate a single common slope plus the same additive effects.

Model structure 2 (m1a-m18a)

The second model structure, model m1a, consisted of the vessel-adjusted CPUE and an odd/even year factor,

$$Y_t = \beta_0 + \beta_1 O_t + \beta_2 \text{CPUE}_t + \epsilon_t. \quad (4)$$

The reference category (i.e., reference intercept) is an even return year. The temperature index for models m2a-m18a was either the SECM survey Icy Strait temperature index (ISTI20_JJ; Murphy et al. 2019) or one of the 16 satellite-derived SST variables (Huang et al. 2017).

Performance Metric: 5-Year One-Step-Ahead MAPE (MAPE5)

The model performance metric MAPE is a standard measure of the accuracy of forecasting and prediction models. It is the average of the absolute percentage differences between predicted and observed values. Because error is expressed as a percentage of the observed value rather than in harvest units, MAPE can be compared across all 54 models; a lower MAPE indicates a more accurate model.

The model summary results for the five-year one-step-ahead MAPE are shown in Table 2 and Table 3; the model with the smallest value is preferred. The one-step-ahead MAPE was calculated as follows:

1. Estimate the regression parameters from data up to year $t - 1$.
2. Predict the log harvest $\hat{Y}_t$ for year t from the predictor variables in year t and the regression parameters estimated in step 1.

3. Back-transform the prediction to the harvest scale with a bias correction and calculate the absolute percentage error (APE) relative to the observed harvest C_t ,

$$\text{APE}_t = \left| \frac{C_t - \exp\left(\hat{Y}_t + \frac{\hat{\sigma}_{t-1}^2}{2}\right)}{C_t} \right|, \quad (5)$$

where $\hat{\sigma}_{t-1}$ is the residual standard error of the model fit to data up to year $t - 1$.

4. Average the absolute percentage errors across the forecast years,

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \text{APE}_t, \quad (6)$$

where n is the number of one-step-ahead forecasts ($n = 5$ for the five-year MAPE).

For example, to calculate the five-year one-step-ahead MAPE for model m1 for the 2027 forecast, the model is fit to data through juvenile year 2020 and used to predict juvenile year 2021 (harvest year 2022), and the absolute percentage error is calculated from that prediction and the observed harvest using Equation 5. The model is then refit to data through juvenile year 2021 to predict juvenile year 2022 (harvest year 2023), and so on, until data through juvenile year 2024 are used to predict juvenile year 2025 (harvest year 2026). The five absolute percentage errors are averaged using Equation 6 to give the five-year one-step-ahead MAPE for model m1. The bias-corrected prediction was used in this calculation so that model performance was evaluated on the same quantity reported as the forecast.

Performance Metric: Leave-One-Out Cross-Validation MAPE (LOOCV)

Model performance was also evaluated with leave-one-out cross-validation (LOOCV). For each year t in the time series, the model was refit to the remaining $n - 1$ years, and the withheld year's harvest was predicted from that fit. Predictions were back-transformed to the harvest scale with a bias correction,

$$\hat{C}_t = \exp\left(\hat{Y}_{-t} + \frac{\hat{\sigma}_{-t}^2}{2}\right), \quad (7)$$

where $\hat{Y}_{-t}$ is the predicted log harvest for year t and $\hat{\sigma}_{-t}$ is the residual standard error of the model fit without year t . The LOOCV MAPE was then calculated as,

$$\text{MAPE}_{\text{LOOCV}} = \frac{100}{n} \sum_{t=1}^n \frac{|C_t - \hat{C}_t|}{C_t}, \quad (8)$$

where C_t is the observed harvest in year t and n is the number of years with observed harvest (27 for the models in structure 1 and 29 for the models in structure 2). Unlike the one-step-ahead MAPE, which forecasts each of the five most recent years from earlier years only, LOOCV uses all remaining years, including later ones, to predict each withheld year. It therefore measures predictive fit across the full time series rather than forward forecasting performance.

Akaike Information Criterion Corrected for Small Sample Sizes (AICc)

Candidate models were compared with Akaike's information criterion corrected for small sample sizes (AICc),

$$\text{AICc} = -2 \ln(\hat{L}) + 2K + \frac{2K(K+1)}{n-K-1}, \quad (9)$$

where $\hat{L}$ is the maximized likelihood, K is the number of estimated parameters (including the residual variance), and n is the number of years in the model data set. AICc balances goodness of fit against model complexity: lower values are preferred, and models with more parameters are penalized. Models were ranked by $\Delta_i = \text{AICc}_i - \text{AICc}_{\min}$, the difference between the AICc of model i and that of the best model in the comparison (Table 2 and Table 3; dAICc). Models with $\Delta_i \leq 2$ have substantial support and are essentially indistinguishable; when they differ in the number of parameters, the most parsimonious is preferred. Models with Δ_i of 4-7 have considerably less support, and those with $\Delta_i > 10$ have essentially no support (Burnham and Anderson 2002).

AICc values are only comparable among models fit to the same response data. Models m1-m18 and m1i-m18i exclude the 2008 (Steller) and 2009 (Chellissa) survey years and were fit to 27 years, so these 36 models were compared with each other (Table 2). Models m1a-m18a include those two years and were fit to 29 years, so they were compared only among themselves (Table 3).

Adjusted R-squared

The adjusted R-squared is a modified version of the R-squared value that adjusts for the number of predictors in the model, balancing model fit against complexity. A higher adjusted R-squared value is preferred. Because the adjusted R-squared depends on the sample size and on the variance of the response variable in the data set, it can only be compared among models fit to the same response data. The 36 models m1-m18 and m1i-m18i share the same 27 years and can be compared with each other (Table 2), but not with models m1a-m18a, which were fit to 29 years (Table 3).

Results

Based upon the 5-year MAPE, the AICc values, significant parameters in the models, and the adjusted R-squared values, model m13a (vessel-adjusted CPUE, an odd/even year factor, and NSEAK_SST_AMJ; Figure 1; Appendix C) was selected. The 2027 forecast using this model has a point estimate of 66.7 million fish (80% prediction interval: 41.5 to 97.5 million fish).

Adding vessel-specific CPUE slopes (i.e., models with an interaction term; m1i-m18i) did not improve the models. For example, for model m5i relative to m5, neither interaction term was significant (Medeia: $p = 0.27$; NW Explorer: $p = 0.33$), an F -test of the two terms together was not significant ($F = 1.47$, $p = 0.25$), and AICc increased by 4.8 units. Adjusted R-squared increased slightly (0.78 vs. 0.77), but this did not offset the added parameters. Across all 18 temperature variables, each interaction model had a higher AICc than its additive counterpart (Table 2).

Table 1: Summary of the data used for the 2027 forecast. The Steller and Chellissa vessels were only used for one year each, 2008 and 2009, respectively, and so those two years are not used in the models that incorporated the adj_raw_pink_log variable. The CPUE variable is the vessel-adjusted CPUE, JYear is the juvenile year, and Year is the harvest year.

JYear	Year	Harvest	odd_even_factor	vessel	adj_raw_pink_log	CPUE
1997	1998	42.4	even	Cobb	2.48	2.48
1998	1999	77.8	odd	Cobb	5.62	5.62
1999	2000	20.2	even	Cobb	1.60	1.60
2000	2001	67.0	odd	Cobb	3.73	3.73
2001	2002	45.3	even	Cobb	2.87	2.87
2002	2003	52.5	odd	Cobb	2.78	2.78
2003	2004	45.3	even	Cobb	3.08	3.08
2004	2005	59.1	odd	Cobb	3.90	3.90
2005	2006	11.6	even	Cobb	2.04	2.04
2006	2007	44.8	odd	Cobb	2.57	2.57
2007	2008	15.9	even	Cobb	1.17	1.17
2008	2009	38.0	odd	Steller	2.45	2.32
2009	2010	24.1	even	Chellissa	3.16	2.33
2010	2011	58.9	odd	NW Explorer	5.57	4.11
2011	2012	21.3	even	NW Explorer	1.97	1.45
2012	2013	94.7	odd	NW Explorer	4.78	3.52
2013	2014	37.2	even	NW Explorer	2.91	2.14
2014	2015	35.1	odd	NW Explorer	5.18	3.82
2015	2016	18.4	even	NW Explorer	3.33	2.45
2016	2017	34.7	odd	NW Explorer	5.90	4.35

Table 1: Summary of the data used for the 2027 forecast. The Steller and Chellissa vessels were only used for one year each, 2008 and 2009, respectively, and so those two years are not used in the models that incorporated the adj_raw_pink_log variable. The CPUE variable is the vessel-adjusted CPUE, JYear is the juvenile year, and Year is the harvest year.

JYear	Year	Harvest	odd_even_factor	vessel	adj_raw_pink_log	CPUE
2017	2018	8.1	even	NW Explorer	0.47	0.35
2018	2019	21.1	odd	Medeia	1.39	1.17
2019	2020	8.1	even	Medeia	1.36	1.14
2020	2021	48.5	odd	Medeia	2.56	2.15
2021	2022	18.3	even	Medeia	1.04	0.87
2022	2023	48.5	odd	Medeia	1.72	1.45
2023	2024	20.1	even	Medeia	1.42	1.19
2024	2025	21.4	odd	Medeia	1.98	1.66
2025	2026	20.4	even	Medeia	1.53	1.29
2026	2027	NA	odd	Medeia	4.18	3.51

Table 2: Summary of the performance metrics for the 36 regression models that use the uncalibrated juvenile index (adj_raw_pink_log) with a vessel factor. Models ending in i include a vessel $\times$ index interaction. Fit is the bias-corrected 2027 harvest forecast (millions of fish) and 80% PI is its prediction interval. dAICc is the difference in AICc from the best of the 36 models. MAPE5 is the five-year one-step-ahead MAPE (%) and LOOCV is the leave-one-out cross-validation MAPE (%).

Model	Temp	Fit	80% PI	AdjR2	dAICc	MAPE5	LOOCV
m7	Icy Strait May	111.7	63.1-179.1	0.77	0.6	20.6	30.8
m3	Chatham May	101.9	58.7-160.5	0.78	0.0	22.3	31.4
m5	Chatham AMJ	90.6	52.4-141.9	0.77	0.7	25.2	33.2
m15	SEAK May	83.3	47.9-130.6	0.76	1.8	25.5	33.0
m11	NSEAK May	95.8	55.5-150.2	0.78	0.1	26.2	31.4
m17	SEAK AMJ	77.2	44.3-121.0	0.76	2.5	28.0	34.3
m13	NSEAK AMJ	91.6	52.8-144.0	0.77	0.9	28.3	33.2
m9	Icy Strait AMJ	107.4	60.3-172.8	0.77	1.5	30.0	33.8
m1	None	56.5	27.3-97.5	0.59	14.3	30.0	46.7
m3i	Chatham May	401.2	140.9-1050.7	0.81	2.1	32.9	33.5
m6	Chatham AMJJ	92.7	50.5-151.5	0.73	5.1	33.3	36.8
m2	ISTI20_JJ	90.6	48.0-150.7	0.71	7.2	33.3	38.6
m18	SEAK AMJJ	82.1	45.0-132.8	0.73	5.7	33.4	36.5
m7i	Icy Strait May	439.8	147.7-1198.8	0.80	3.8	34.4	34.7
m2i	ISTI20_JJ	207.9	58.2-652.5	0.70	14.1	34.9	46.6
m5i	Chatham AMJ	237.2	80.7-633.8	0.78	5.5	35.0	36.3
m14	NSEAK AMJJ	95.6	52.1-156.5	0.73	4.9	35.7	36.3
m15i	SEAK May	314.4	108.2-835.1	0.79	4.0	36.3	35.3
m11i	NSEAK May	386.2	137.1-1002.0	0.81	1.8	36.6	33.6
m10	Icy Strait AMJJ	101.5	53.7-169.8	0.72	6.1	37.0	38.0
m16	SEAK MJJ	76.6	40.3-127.1	0.69	9.2	37.7	39.7
m4	Chatham MJJ	86.2	44.2-146.4	0.68	9.7	37.8	40.7
m17i	SEAK AMJ	192.3	64.1-521.9	0.77	7.0	38.1	37.6
m13i	NSEAK AMJ	249.8	85.4-665.4	0.79	5.1	38.9	36.3
m1i	None	112.5	24.9-421.3	0.57	20.9	39.9	54.1
m9i	Icy Strait AMJ	287.3	93.1-802.1	0.77	7.0	40.6	39.5
m12	NSEAK MJJ	87.0	45.3-146.4	0.69	8.8	40.7	39.9
m8	Icy Strait MJJ	88.6	45.2-151.3	0.68	9.8	40.9	41.7
m6i	Chatham AMJJ	176.2	53.2-518.1	0.73	11.6	41.8	42.1
m18i	SEAK AMJJ	148.3	44.9-434.6	0.73	11.8	41.9	41.7
m14i	NSEAK AMJJ	190.4	58.4-553.6	0.74	10.7	45.0	41.3
m16i	SEAK MJJ	129.8	35.5-412.6	0.68	16.2	46.5	46.8
m10i	Icy Strait AMJJ	215.8	62.5-658.4	0.72	12.6	46.7	45.6
m4i	Chatham MJJ	159.3	42.4-517.8	0.67	16.7	47.1	48.5

Table 2: Summary of the performance metrics for the 36 regression models that use the uncalibrated juvenile index (adj_raw_pink_log) with a vessel factor. Models ending in i include a vessel $\times$ index interaction. Fit is the bias-corrected 2027 harvest forecast (millions of fish) and 80% PI is its prediction interval. dAICc is the difference in AICc from the best of the 36 models. MAPE5 is the five-year one-step-ahead MAPE (%) and LOOCV is the leave-one-out cross-validation MAPE (%).

Model	Temp	Fit	80% PI	AdjR2	dAICc	MAPE5	LOOCV
m12i	NSEAK MJJ	164.7	45.3-521.6	0.69	15.4	50.7	46.6
m8i	Icy Strait MJJ	184.3	48.4-607.2	0.67	16.8	51.4	50.8

Table 3: Summary of the performance metrics for the 18 regression models with vessel-adjusted CPUE. Fit is the bias-corrected 2027 harvest forecast (millions of fish) and 80% PI is its prediction interval. dAICc is the difference in AICc from the best of these 18 models; these models were fit to 29 years of data, so their AICc values are not comparable with the models in Table 2. MAPE5 is the five-year one-step-ahead MAPE (%) and LOOCV is the leave-one-out cross-validation MAPE (%).

Model	Temp	Fit	80% PI	AdjR2	dAICc	MAPE5	LOOCV
m5a	Chatham AMJ	69.5	43.5-101.3	0.77	0.8	20.5	29.6
m9a	Icy Strait AMJ	80.4	49.4-119	0.77	1.6	21.2	29.4
m2a	ISTI20_JJ	67.5	38.6-104	0.69	9.8	21.2	36.3
m13a	NSEAK AMJ	66.7	41.5-97.5	0.77	1.7	22.4	29.8
m3a	Chatham May	74.7	46.9-108.7	0.78	0.0	23.3	28.8
m11a	NSEAK May	69.2	43.4-100.6	0.77	0.5	23.6	29.2
m7a	Icy Strait May	80.6	49.9-118.6	0.77	0.9	24.1	29.1
m15a	SEAK May	60.6	37.5-88.6	0.76	2.4	24.6	30.9
m17a	SEAK AMJ	57.2	35.2-84.1	0.75	3.4	24.8	31.2
m1a	None	56.8	30.7-90	0.62	13.9	25.3	39.7
m10a	Icy Strait AMJJ	79.5	46.7-121.2	0.73	6.0	25.5	33.7
m6a	Chatham AMJJ	70.1	41.8-105.2	0.73	5.6	25.8	33.1
m14a	NSEAK AMJJ	67.7	40.3-101.7	0.73	6.1	27.1	33.2
m18a	SEAK AMJJ	59.0	35-88.5	0.72	7.0	29.1	34.0
m8a	Icy Strait MJJ	75.4	43-116.9	0.70	9.0	29.8	36.3
m4a	Chatham MJJ	69.3	39.8-106.5	0.70	9.2	30.0	36.1
m12a	NSEAK MJJ	65.5	38-100	0.70	8.9	30.7	35.4
m16a	SEAK MJJ	58.1	33.7-88.4	0.70	9.3	31.4	35.6

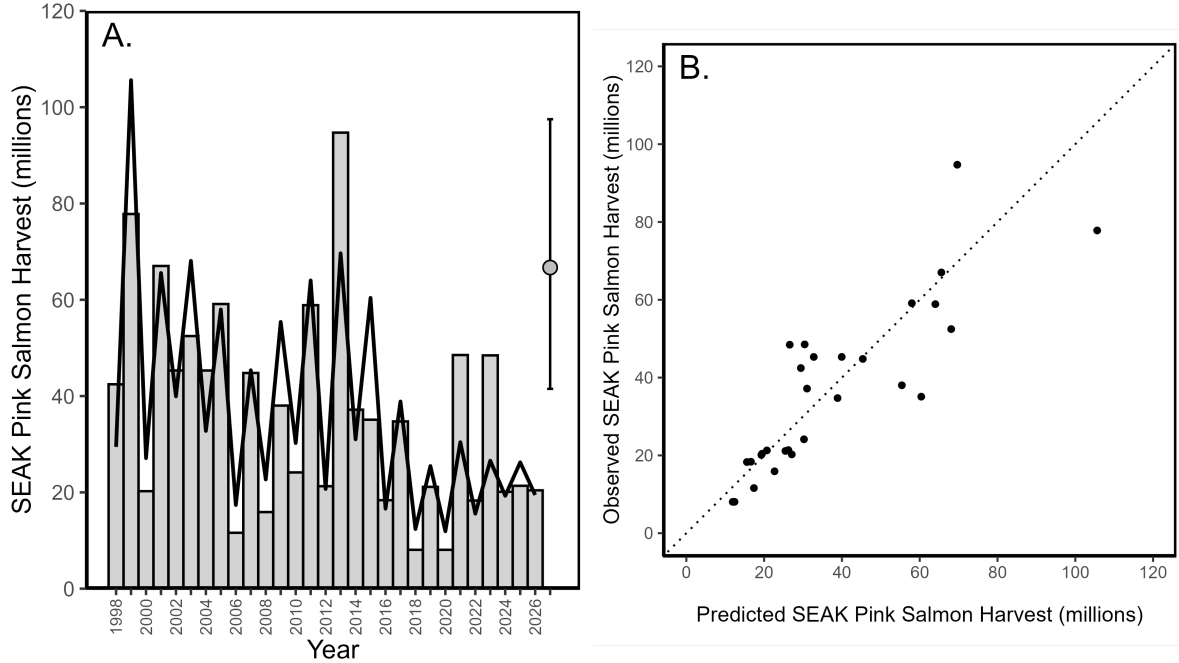


Figure 1: A. SEAK pink salmon harvest (millions) by year with the model fit (line) from the selected model (m13a). The 2027 forecast (66.7 million fish) is shown as a grey circle with its 80% prediction interval (41.5-97.5 million fish). B. Observed SEAK pink salmon harvest (millions) against the bias-corrected fitted values from model m13a. The dotted line is a one-to-one reference line.

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Appendix A

Variable definitions

adj_raw_pink_log: First, the raw juvenile pink salmon catch by year, month, haul, and area was adjusted to a 20-minute haul (raw_pink_catch). Then, these values were log-transformed (i.e., $\ln(\text{raw_pink_catch} + 1)$). Next, the maximum value in either June or July, whichever month had the highest in a given year, was used as the adj_raw_pink_log value for the year (i.e., $\text{adj_raw_pink_log} = \text{maximum}(\ln(\text{raw_pink_catch} + 1))$ in either June or July).

vessel-adjusted CPUE: The average $\ln(\text{CPUE} + 1)$ for juvenile pink salmon catches in either June or July, whichever month had the highest average in a given year, where effort was a standard trawl haul (20 minutes). The juvenile CPUE data were adjusted using vessel calibration factors to account for differences in fishing power among vessels.

ISTI20_JJ: The average 20-m integrated water column temperature at the eight stations in Icy Strait (Icy Strait and Upper Chatham transects) sampled during the SECM surveys in June and July of each year (in degrees Celsius). There was no survey in May in 2025 or 2026. Therefore, this temperature index no longer includes May in the time series.

Satellite SST variables

Icy_Strait_SST_May: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in May.

Icy_Strait_SST_MJJ: The Icy Strait region (as above). This variable is the average SST in May through July.

Icy_Strait_SST_AMJ: The Icy Strait region (as above). This variable is the average SST in April through June.

Icy_Strait_SST_AMJJ: The Icy Strait region (as above). This variable is the average SST in April through July.

Chatham_SST_May: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, and south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in May.

Chatham_SST_MJJ: The Chatham and Icy Straits region (as above). This variable is the average SST in May through July.

Chatham_SST_AMJ: The Chatham and Icy Straits region (as above). This variable is the average SST in April through June. The last time this variable was incorporated in the forecasting process was the 2023 forecast.

Chatham_SST_AMJJ: The Chatham and Icy Straits region (as above). This variable is the average SST in April through July.

NSEAK_SST_May: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in May.

NSEAK_SST_MJJ: The NSEAK region (as above). This variable is the average SST in May through July.

NSEAK_SST_AMJ: The NSEAK region (as above). This variable is the average SST in April through June.

NSEAK_SST_AMJJ: The NSEAK region (as above). This variable is the average SST in April through July.

SEAK_SST_May: The SEAK region encompasses Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in May. The last time this variable was incorporated in the forecasting process was the 2023 forecast.

SEAK_SST_MJJ: The SEAK region (as above). This variable is the average SST in May through July.

SEAK_SST_AMJ: The SEAK region (as above). This variable is the average SST in April through June.

SEAK_SST_AMJJ: The SEAK region (as above). This variable is the average SST in April through July.

Appendix B

Table 4: Parameter estimates for the 18 individual models m1i-m18i.

Model	Term	Estimate	Standard Error	Statistic	p value
m1i	(Intercept)	2.574	0.362	7.114	0.000
m1i	as.factor(vessel)Medeia	-0.493	0.694	-0.711	0.485
m1i	as.factor(vessel)NW Explorer	-0.144	0.501	-0.288	0.776
m1i	adj_raw_pink_log	0.311	0.132	2.362	0.028
m1i	as.factor(odd_even_factor)odd	0.358	0.245	1.463	0.159
m1i	as.factor(vessel)Medeia:adj_raw_pink_log	0.213	0.387	0.551	0.588
m1i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.094	0.142	-0.664	0.515
m2i	(Intercept)	7.633	1.634	4.672	0.000
m2i	as.factor(vessel)Medeia	-0.518	0.577	-0.898	0.381
m2i	as.factor(vessel)NW Explorer	-0.482	0.430	-1.120	0.277
m2i	adj_raw_pink_log	0.454	0.119	3.830	0.001
m2i	ISTI20_JJ	-0.526	0.167	-3.151	0.005
m2i	as.factor(odd_even_factor)odd	-0.010	0.235	-0.042	0.967
m2i	as.factor(vessel)Medeia:adj_raw_pink_log	0.294	0.323	0.911	0.374
m2i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.056	0.118	-0.476	0.640
m3i	(Intercept)	6.305	0.770	8.193	0.000
m3i	as.factor(vessel)Medeia	-0.536	0.462	-1.160	0.261
m3i	as.factor(vessel)NW Explorer	0.004	0.335	0.012	0.991
m3i	adj_raw_pink_log	0.465	0.093	5.009	0.000
m3i	Chatham_SST_May	-0.552	0.108	-5.105	0.000
m3i	as.factor(odd_even_factor)odd	0.113	0.170	0.667	0.513
m3i	as.factor(vessel)Medeia:adj_raw_pink_log	0.467	0.262	1.778	0.091
m3i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.096	0.094	-1.018	0.321
m4i	(Intercept)	6.512	1.499	4.345	0.000
m4i	as.factor(vessel)Medeia	-0.246	0.613	-0.401	0.693
m4i	as.factor(vessel)NW Explorer	-0.096	0.438	-0.219	0.829
m4i	adj_raw_pink_log	0.386	0.118	3.261	0.004
m4i	Chatham_SST_MJJ	-0.422	0.157	-2.688	0.015
m4i	as.factor(odd_even_factor)odd	0.255	0.217	1.173	0.255
m4i	as.factor(vessel)Medeia:adj_raw_pink_log	0.180	0.338	0.531	0.601
m4i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.101	0.124	-0.816	0.424
m5i	(Intercept)	6.911	0.985	7.019	0.000
m5i	as.factor(vessel)Medeia	-0.357	0.493	-0.726	0.477
m5i	as.factor(vessel)NW Explorer	0.001	0.356	0.003	0.998
m5i	adj_raw_pink_log	0.440	0.098	4.515	0.000
m5i	Chatham_SST_AMJ	-0.612	0.134	-4.563	0.000

Table 4: Parameter estimates for the 18 individual models m1i-m18i.

Model	Term	Estimate	Standard Error	Statistic	p value
m5i	as.factor(odd_even_factor)odd	0.149	0.180	0.832	0.416
m5i	as.factor(vessel)Medeia:adj_raw_pink_log	0.314	0.275	1.141	0.268
m5i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.101	0.100	-1.007	0.327
m6i	(Intercept)	7.107	1.299	5.469	0.000
m6i	as.factor(vessel)Medeia	-0.180	0.557	-0.323	0.750
m6i	as.factor(vessel)NW Explorer	-0.054	0.398	-0.136	0.893
m6i	adj_raw_pink_log	0.422	0.109	3.876	0.001
m6i	Chatham_SST_AMJJ	-0.553	0.155	-3.577	0.002
m6i	as.factor(odd_even_factor)odd	0.201	0.199	1.007	0.326
m6i	as.factor(vessel)Medeia:adj_raw_pink_log	0.183	0.307	0.595	0.559
m6i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.112	0.112	-0.993	0.333
m7i	(Intercept)	5.975	0.745	8.017	0.000
m7i	as.factor(vessel)Medeia	-0.606	0.477	-1.270	0.219
m7i	as.factor(vessel)NW Explorer	-0.099	0.344	-0.287	0.777
m7i	adj_raw_pink_log	0.448	0.095	4.729	0.000
m7i	Icy_Strait_SST_May	-0.531	0.110	-4.841	0.000
m7i	as.factor(odd_even_factor)odd	0.167	0.173	0.965	0.346
m7i	as.factor(vessel)Medeia:adj_raw_pink_log	0.487	0.271	1.793	0.089
m7i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.062	0.097	-0.637	0.532
m8i	(Intercept)	6.045	1.333	4.536	0.000
m8i	as.factor(vessel)Medeia	-0.420	0.607	-0.692	0.497
m8i	as.factor(vessel)NW Explorer	-0.162	0.438	-0.371	0.715
m8i	adj_raw_pink_log	0.367	0.117	3.136	0.005
m8i	Icy_Strait_SST_MJJ	-0.361	0.135	-2.681	0.015
m8i	as.factor(odd_even_factor)odd	0.291	0.216	1.351	0.192
m8i	as.factor(vessel)Medeia:adj_raw_pink_log	0.237	0.338	0.702	0.491
m8i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.084	0.124	-0.681	0.504
m9i	(Intercept)	6.609	0.973	6.794	0.000
m9i	as.factor(vessel)Medeia	-0.441	0.506	-0.872	0.394
m9i	as.factor(vessel)NW Explorer	-0.105	0.365	-0.288	0.777
m9i	adj_raw_pink_log	0.418	0.099	4.216	0.000
m9i	Icy_Strait_SST_AMJ	-0.586	0.136	-4.310	0.000
m9i	as.factor(odd_even_factor)odd	0.228	0.181	1.256	0.224
m9i	as.factor(vessel)Medeia:adj_raw_pink_log	0.343	0.284	1.208	0.242
m9i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.068	0.103	-0.662	0.516
m10i	(Intercept)	6.523	1.198	5.444	0.000
m10i	as.factor(vessel)Medeia	-0.344	0.563	-0.610	0.549
m10i	as.factor(vessel)NW Explorer	-0.127	0.405	-0.314	0.757
m10i	adj_raw_pink_log	0.403	0.110	3.666	0.002

Table 4: Parameter estimates for the 18 individual models m1i-m18i.

Model	Term	Estimate	Standard Error	Statistic	p value
m10i	Icy_Strait_SST_AMJJ	-0.480	0.141	-3.399	0.003
m10i	as.factor(odd_even_factor)odd	0.254	0.201	1.267	0.221
m10i	as.factor(vessel)Medeia:adj_raw_pink_log	0.238	0.313	0.760	0.457
m10i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.095	0.115	-0.830	0.417
m11i	(Intercept)	6.018	0.709	8.486	0.000
m11i	as.factor(vessel)Medeia	-0.479	0.459	-1.044	0.310
m11i	as.factor(vessel)NW Explorer	0.072	0.334	0.215	0.832
m11i	adj_raw_pink_log	0.446	0.091	4.901	0.000
m11i	NSEAK_SST_May	-0.516	0.100	-5.161	0.000
m11i	as.factor(odd_even_factor)odd	0.156	0.167	0.938	0.360
m11i	as.factor(vessel)Medeia:adj_raw_pink_log	0.469	0.261	1.800	0.088
m11i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.107	0.094	-1.136	0.270
m12i	(Intercept)	6.514	1.383	4.710	0.000
m12i	as.factor(vessel)Medeia	-0.132	0.604	-0.219	0.829
m12i	as.factor(vessel)NW Explorer	-0.011	0.429	-0.027	0.979
m12i	adj_raw_pink_log	0.390	0.115	3.379	0.003
m12i	NSEAK_SST_MJJ	-0.425	0.145	-2.923	0.009
m12i	as.factor(odd_even_factor)odd	0.264	0.211	1.252	0.226
m12i	as.factor(vessel)Medeia:adj_raw_pink_log	0.178	0.330	0.539	0.596
m12i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.115	0.121	-0.950	0.354
m13i	(Intercept)	6.733	0.935	7.197	0.000
m13i	as.factor(vessel)Medeia	-0.259	0.491	-0.528	0.604
m13i	as.factor(vessel)NW Explorer	0.096	0.356	0.268	0.791
m13i	adj_raw_pink_log	0.435	0.096	4.504	0.000
m13i	NSEAK_SST_AMJ	-0.602	0.130	-4.621	0.000
m13i	as.factor(odd_even_factor)odd	0.184	0.177	1.042	0.310
m13i	as.factor(vessel)Medeia:adj_raw_pink_log	0.317	0.273	1.162	0.260
m13i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.117	0.100	-1.168	0.257
m14i	(Intercept)	7.003	1.226	5.711	0.000
m14i	as.factor(vessel)Medeia	-0.084	0.553	-0.151	0.881
m14i	as.factor(vessel)NW Explorer	0.027	0.394	0.069	0.946
m14i	adj_raw_pink_log	0.422	0.107	3.943	0.001
m14i	NSEAK_SST_AMJJ	-0.548	0.148	-3.712	0.001
m14i	as.factor(odd_even_factor)odd	0.224	0.195	1.152	0.263
m14i	as.factor(vessel)Medeia:adj_raw_pink_log	0.188	0.302	0.623	0.541
m14i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.127	0.111	-1.142	0.268
m15i	(Intercept)	6.120	0.780	7.848	0.000
m15i	as.factor(vessel)Medeia	-0.388	0.479	-0.809	0.428
m15i	as.factor(vessel)NW Explorer	0.115	0.350	0.330	0.745

Table 4: Parameter estimates for the 18 individual models m1i-m18i.

Model	Term	Estimate	Standard Error	Statistic	p value
m15i	adj_raw_pink_log	0.460	0.096	4.791	0.000
m15i	SEAK_SST_May	-0.495	0.103	-4.800	0.000
m15i	as.factor(odd_even_factor)odd	0.145	0.175	0.830	0.417
m15i	as.factor(vessel)Medeia:adj_raw_pink_log	0.439	0.271	1.619	0.122
m15i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.121	0.098	-1.238	0.231
m16i	(Intercept)	6.423	1.414	4.541	0.000
m16i	as.factor(vessel)Medeia	-0.015	0.623	-0.023	0.982
m16i	as.factor(vessel)NW Explorer	0.019	0.437	0.043	0.966
m16i	adj_raw_pink_log	0.389	0.117	3.318	0.004
m16i	SEAK_SST_MJJ	-0.397	0.142	-2.791	0.012
m16i	as.factor(odd_even_factor)odd	0.260	0.215	1.212	0.240
m16i	as.factor(vessel)Medeia:adj_raw_pink_log	0.136	0.335	0.404	0.690
m16i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.114	0.123	-0.933	0.362
m17i	(Intercept)	6.745	1.002	6.733	0.000
m17i	as.factor(vessel)Medeia	-0.150	0.512	-0.293	0.773
m17i	as.factor(vessel)NW Explorer	0.132	0.371	0.357	0.725
m17i	adj_raw_pink_log	0.445	0.101	4.412	0.000
m17i	SEAK_SST_AMJ	-0.564	0.131	-4.316	0.000
m17i	as.factor(odd_even_factor)odd	0.180	0.183	0.981	0.339
m17i	as.factor(vessel)Medeia:adj_raw_pink_log	0.275	0.282	0.972	0.343
m17i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.132	0.104	-1.270	0.220
m18i	(Intercept)	6.971	1.277	5.457	0.000
m18i	as.factor(vessel)Medeia	0.034	0.573	0.060	0.953
m18i	as.factor(vessel)NW Explorer	0.070	0.404	0.172	0.865
m18i	adj_raw_pink_log	0.422	0.110	3.852	0.001
m18i	SEAK_SST_AMJJ	-0.515	0.146	-3.533	0.002
m18i	as.factor(odd_even_factor)odd	0.225	0.199	1.132	0.272
m18i	as.factor(vessel)Medeia:adj_raw_pink_log	0.150	0.309	0.484	0.634
m18i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.129	0.113	-1.141	0.268

Table 5: Parameter estimates for the 18 individual models m1-m18.

Model	Term	Estimate	Standard Error	Statistic	p value
m1	(Intercept)	2.740	0.241	11.379	0.000
m1	as.factor(odd_even_factor)odd	0.423	0.226	1.870	0.075
m1	as.factor(vessel)Medeia	-0.236	0.232	-1.019	0.319
m1	as.factor(vessel)NW Explorer	-0.442	0.209	-2.109	0.047
m1	adj_raw_pink_log	0.243	0.090	2.697	0.013
m2	(Intercept)	7.707	1.559	4.943	0.000
m2	ISTI20_JJ	-0.524	0.163	-3.212	0.004
m2	as.factor(odd_even_factor)odd	0.065	0.220	0.295	0.771
m2	as.factor(vessel)Medeia	-0.100	0.199	-0.503	0.620
m2	as.factor(vessel)NW Explorer	-0.658	0.188	-3.498	0.002
m2	adj_raw_pink_log	0.409	0.092	4.468	0.000
m3	(Intercept)	6.174	0.789	7.830	0.000
m3	Chatham_SST_May	-0.509	0.114	-4.468	0.000
m3	as.factor(odd_even_factor)odd	0.246	0.170	1.444	0.163
m3	as.factor(vessel)Medeia	0.094	0.185	0.509	0.616
m3	as.factor(vessel)NW Explorer	-0.310	0.156	-1.980	0.061
m3	adj_raw_pink_log	0.379	0.073	5.206	0.000
m4	(Intercept)	6.703	1.456	4.603	0.000
m4	Chatham_SST_MJJ	-0.423	0.154	-2.750	0.012
m4	as.factor(odd_even_factor)odd	0.314	0.202	1.551	0.136
m4	as.factor(vessel)Medeia	-0.047	0.214	-0.219	0.829
m4	as.factor(vessel)NW Explorer	-0.416	0.184	-2.262	0.034
m4	adj_raw_pink_log	0.314	0.083	3.774	0.001
m5	(Intercept)	6.960	0.987	7.054	0.000
m5	Chatham_SST_AMJ	-0.594	0.137	-4.349	0.000
m5	as.factor(odd_even_factor)odd	0.242	0.173	1.399	0.177
m5	as.factor(vessel)Medeia	0.045	0.184	0.247	0.807
m5	as.factor(vessel)NW Explorer	-0.322	0.158	-2.040	0.054
m5	adj_raw_pink_log	0.362	0.072	5.006	0.000
m6	(Intercept)	7.271	1.280	5.680	0.000
m6	Chatham_SST_AMJJ	-0.549	0.153	-3.581	0.002
m6	as.factor(odd_even_factor)odd	0.264	0.188	1.407	0.174
m6	as.factor(vessel)Medeia	0.012	0.199	0.058	0.954
m6	as.factor(vessel)NW Explorer	-0.409	0.169	-2.417	0.025
m6	adj_raw_pink_log	0.343	0.078	4.401	0.000
m7	(Intercept)	5.832	0.732	7.965	0.000
m7	Icy_Strait_SST_May	-0.492	0.113	-4.354	0.000
m7	as.factor(odd_even_factor)odd	0.291	0.170	1.710	0.102
m7	as.factor(vessel)Medeia	0.089	0.187	0.474	0.641

Table 5: Parameter estimates for the 18 individual models m1-m18.

Model	Term	Estimate	Standard Error	Statistic	p value
m7	as.factor(vessel)NW Explorer	-0.303	0.159	-1.912	0.070
m7	adj_raw_pink_log	0.385	0.074	5.171	0.000
m8	(Intercept)	6.195	1.280	4.839	0.000
m8	Icy_Strait_SST_MJJ	-0.361	0.132	-2.736	0.012
m8	as.factor(odd_even_factor)odd	0.359	0.200	1.793	0.087
m8	as.factor(vessel)Medeia	-0.116	0.208	-0.557	0.584
m8	as.factor(vessel)NW Explorer	-0.429	0.184	-2.330	0.030
m8	adj_raw_pink_log	0.305	0.082	3.700	0.001
m9	(Intercept)	6.636	0.944	7.028	0.000
m9	Icy_Strait_SST_AMJ	-0.573	0.136	-4.204	0.000
m9	as.factor(odd_even_factor)odd	0.316	0.172	1.833	0.081
m9	as.factor(vessel)Medeia	0.038	0.186	0.204	0.840
m9	as.factor(vessel)NW Explorer	-0.323	0.161	-2.014	0.057
m9	adj_raw_pink_log	0.363	0.074	4.917	0.000
m10	(Intercept)	6.661	1.168	5.702	0.000
m10	Icy_Strait_SST_AMJJ	-0.476	0.140	-3.406	0.003
m10	as.factor(odd_even_factor)odd	0.325	0.188	1.727	0.099
m10	as.factor(vessel)Medeia	-0.049	0.198	-0.248	0.806
m10	as.factor(vessel)NW Explorer	-0.428	0.172	-2.488	0.021
m10	adj_raw_pink_log	0.333	0.079	4.235	0.000
m11	(Intercept)	5.901	0.733	8.056	0.000
m11	NSEAK_SST_May	-0.472	0.106	-4.447	0.000
m11	as.factor(odd_even_factor)odd	0.289	0.169	1.715	0.101
m11	as.factor(vessel)Medeia	0.139	0.190	0.733	0.472
m11	as.factor(vessel)NW Explorer	-0.279	0.158	-1.764	0.092
m11	adj_raw_pink_log	0.354	0.071	5.001	0.000
m12	(Intercept)	6.669	1.358	4.910	0.000
m12	NSEAK_SST_MJJ	-0.419	0.143	-2.927	0.008
m12	as.factor(odd_even_factor)odd	0.327	0.198	1.655	0.113
m12	as.factor(vessel)Medeia	0.047	0.222	0.213	0.833
m12	as.factor(vessel)NW Explorer	-0.377	0.182	-2.071	0.051
m12	adj_raw_pink_log	0.308	0.081	3.809	0.001
m13	(Intercept)	6.761	0.951	7.111	0.000
m13	NSEAK_SST_AMJ	-0.576	0.134	-4.307	0.000
m13	as.factor(odd_even_factor)odd	0.281	0.172	1.637	0.116
m13	as.factor(vessel)Medeia	0.129	0.192	0.669	0.511
m13	as.factor(vessel)NW Explorer	-0.281	0.161	-1.746	0.095
m13	adj_raw_pink_log	0.345	0.071	4.837	0.000
m14	(Intercept)	7.126	1.224	5.821	0.000

Table 5: Parameter estimates for the 18 individual models m1-m18.

Model	Term	Estimate	Standard Error	Statistic	p value
m14	NSEAK_SST_AMJJ	-0.535	0.148	-3.628	0.002
m14	as.factor(odd_even_factor)odd	0.294	0.185	1.590	0.127
m14	as.factor(vessel)Medeia	0.098	0.207	0.473	0.641
m14	as.factor(vessel)NW Explorer	-0.377	0.169	-2.228	0.037
m14	adj_raw_pink_log	0.331	0.076	4.338	0.000
m15	(Intercept)	6.006	0.807	7.439	0.000
m15	SEAK_SST_May	-0.451	0.109	-4.153	0.000
m15	as.factor(odd_even_factor)odd	0.277	0.175	1.581	0.129
m15	as.factor(vessel)Medeia	0.165	0.200	0.824	0.419
m15	as.factor(vessel)NW Explorer	-0.283	0.163	-1.729	0.098
m15	adj_raw_pink_log	0.358	0.074	4.850	0.000
m16	(Intercept)	6.615	1.379	4.796	0.000
m16	SEAK_SST_MJJ	-0.395	0.139	-2.842	0.010
m16	as.factor(odd_even_factor)odd	0.313	0.200	1.563	0.133
m16	as.factor(vessel)Medeia	0.101	0.234	0.431	0.671
m16	as.factor(vessel)NW Explorer	-0.345	0.185	-1.863	0.077
m16	adj_raw_pink_log	0.309	0.082	3.774	0.001
m17	(Intercept)	6.787	1.021	6.647	0.000
m17	SEAK_SST_AMJ	-0.539	0.134	-4.030	0.001
m17	as.factor(odd_even_factor)odd	0.272	0.178	1.532	0.140
m17	as.factor(vessel)Medeia	0.154	0.203	0.761	0.455
m17	as.factor(vessel)NW Explorer	-0.292	0.165	-1.765	0.092
m17	adj_raw_pink_log	0.346	0.074	4.684	0.000
m18	(Intercept)	7.115	1.269	5.607	0.000
m18	SEAK_SST_AMJJ	-0.505	0.145	-3.490	0.002
m18	as.factor(odd_even_factor)odd	0.287	0.188	1.524	0.142
m18	as.factor(vessel)Medeia	0.152	0.219	0.695	0.495
m18	as.factor(vessel)NW Explorer	-0.344	0.173	-1.987	0.060
m18	adj_raw_pink_log	0.330	0.078	4.256	0.000

Table 6: Parameter estimates for the 18 individual models m1a-m18a.

Model	Term	Estimate	Standard Error	Statistic	p value
m1a	(Intercept)	2.496	0.163	15.298	0.000
m1a	CPUE	0.304	0.072	4.201	0.000
m1a	as.factor(odd_even_factor)odd	0.400	0.175	2.289	0.030
m2a	(Intercept)	6.028	1.353	4.456	0.000
m2a	CPUE	0.396	0.074	5.343	0.000
m2a	as.factor(odd_even_factor)odd	0.196	0.176	1.112	0.277
m2a	ISTI20_JJ	-0.368	0.140	-2.627	0.015
m3a	(Intercept)	5.528	0.693	7.973	0.000
m3a	CPUE	0.363	0.057	6.400	0.000
m3a	as.factor(odd_even_factor)odd	0.357	0.134	2.670	0.013
m3a	Chatham_SST_May	-0.417	0.094	-4.445	0.000
m4a	(Intercept)	5.832	1.221	4.776	0.000
m4a	CPUE	0.334	0.066	5.099	0.000
m4a	as.factor(odd_even_factor)odd	0.346	0.157	2.199	0.037
m4a	Chatham_SST_MJJ	-0.345	0.125	-2.751	0.011
m5a	(Intercept)	6.125	0.852	7.185	0.000
m5a	CPUE	0.360	0.057	6.269	0.000
m5a	as.factor(odd_even_factor)odd	0.332	0.136	2.442	0.022
m5a	Chatham_SST_AMJ	-0.486	0.113	-4.304	0.000
m6a	(Intercept)	6.217	1.089	5.709	0.000
m6a	CPUE	0.345	0.062	5.577	0.000
m6a	as.factor(odd_even_factor)odd	0.335	0.148	2.260	0.033
m6a	Chatham_SST_AMJJ	-0.434	0.126	-3.445	0.002
m7a	(Intercept)	5.221	0.647	8.067	0.000
m7a	CPUE	0.370	0.058	6.373	0.000
m7a	as.factor(odd_even_factor)odd	0.384	0.135	2.837	0.009
m7a	Icy_Strait_SST_May	-0.399	0.093	-4.293	0.000
m8a	(Intercept)	5.561	1.105	5.032	0.000
m8a	CPUE	0.341	0.066	5.193	0.000
m8a	as.factor(odd_even_factor)odd	0.362	0.156	2.320	0.029
m8a	Icy_Strait_SST_MJJ	-0.313	0.112	-2.798	0.010
m9a	(Intercept)	5.874	0.822	7.144	0.000
m9a	CPUE	0.361	0.058	6.186	0.000
m9a	as.factor(odd_even_factor)odd	0.388	0.137	2.831	0.009
m9a	Icy_Strait_SST_AMJ	-0.471	0.113	-4.159	0.000
m10a	(Intercept)	5.876	1.012	5.806	0.000
m10a	CPUE	0.350	0.063	5.588	0.000
m10a	as.factor(odd_even_factor)odd	0.365	0.148	2.461	0.021
m10a	Icy_Strait_SST_AMJJ	-0.398	0.118	-3.371	0.002

Table 6: Parameter estimates for the 18 individual models m1a-m18a.

Model	Term	Estimate	Standard Error	Statistic	p value
m11a	(Intercept)	5.265	0.648	8.126	0.000
m11a	CPUE	0.334	0.056	5.960	0.000
m11a	as.factor(odd_even_factor)odd	0.393	0.134	2.924	0.007
m11a	NSEAK_SST_May	-0.378	0.087	-4.357	0.000
m12a	(Intercept)	5.629	1.118	5.034	0.000
m12a	CPUE	0.311	0.064	4.836	0.000
m12a	as.factor(odd_even_factor)odd	0.382	0.155	2.459	0.021
m12a	NSEAK_SST_MJJ	-0.318	0.112	-2.826	0.009
m13a	(Intercept)	5.857	0.819	7.153	0.000
m13a	CPUE	0.328	0.057	5.752	0.000
m13a	as.factor(odd_even_factor)odd	0.384	0.137	2.801	0.010
m13a	NSEAK_SST_AMJ	-0.450	0.108	-4.156	0.000
m14a	(Intercept)	5.924	1.029	5.755	0.000
m14a	CPUE	0.318	0.061	5.188	0.000
m14a	as.factor(odd_even_factor)odd	0.380	0.148	2.566	0.017
m14a	NSEAK_SST_AMJJ	-0.397	0.118	-3.361	0.002
m15a	(Intercept)	5.266	0.701	7.512	0.000
m15a	CPUE	0.329	0.058	5.699	0.000
m15a	as.factor(odd_even_factor)odd	0.398	0.139	2.862	0.008
m15a	SEAK_SST_May	-0.349	0.087	-4.021	0.000
m16a	(Intercept)	5.524	1.117	4.948	0.000
m16a	CPUE	0.300	0.065	4.640	0.000
m16a	as.factor(odd_even_factor)odd	0.392	0.156	2.505	0.019
m16a	SEAK_SST_MJJ	-0.290	0.106	-2.736	0.011
m17a	(Intercept)	5.787	0.866	6.686	0.000
m17a	CPUE	0.322	0.059	5.489	0.000
m17a	as.factor(odd_even_factor)odd	0.389	0.141	2.755	0.011
m17a	SEAK_SST_AMJ	-0.408	0.106	-3.847	0.001
m18a	(Intercept)	5.801	1.044	5.558	0.000
m18a	CPUE	0.307	0.062	4.941	0.000
m18a	as.factor(odd_even_factor)odd	0.393	0.150	2.619	0.015
m18a	SEAK_SST_AMJJ	-0.358	0.112	-3.195	0.004

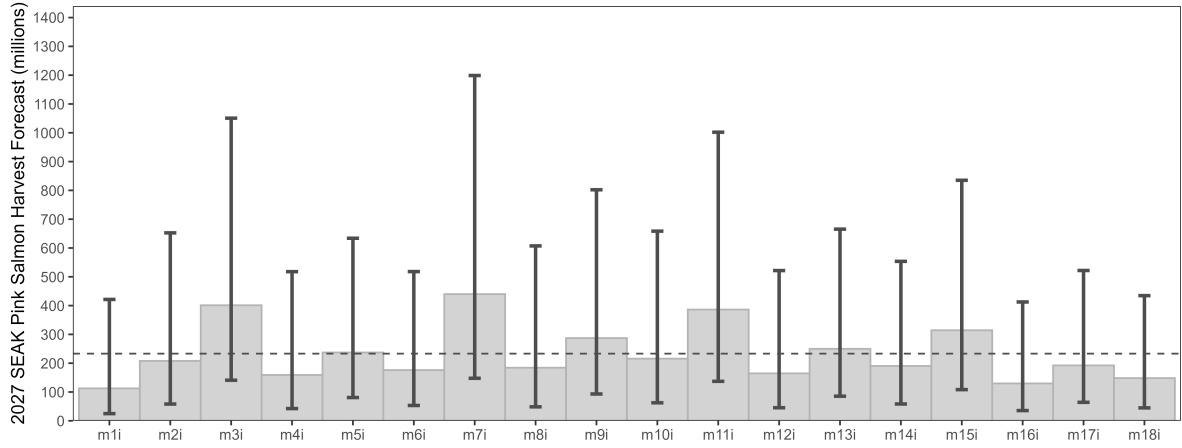


Figure 2: Bias-corrected forecasts (grey bars) for the 18 regression models m1i-m18i with 80% prediction intervals (vertical lines). A horizontal dashed line at the average of the model forecasts is shown for reference only.

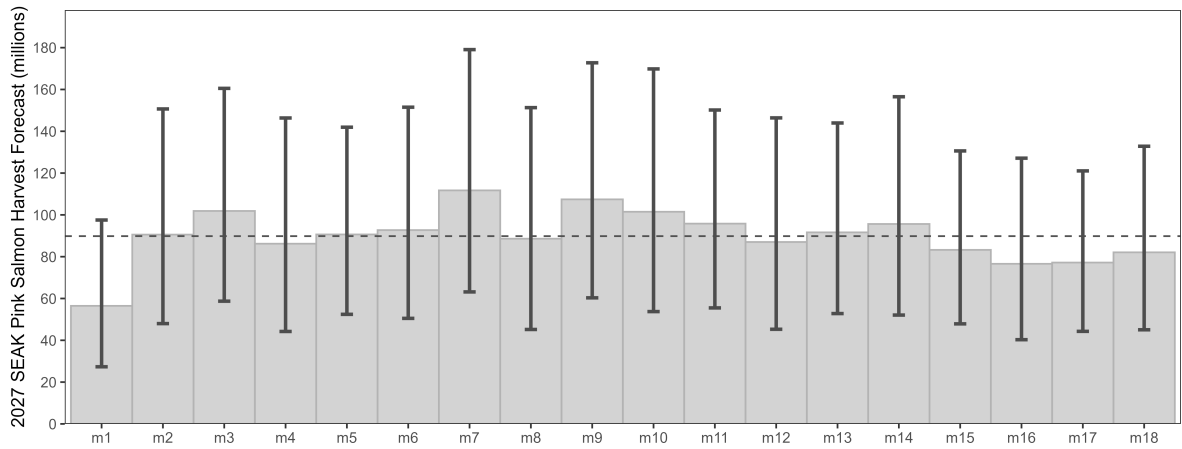


Figure 3: Bias-corrected forecasts (grey bars) for the 18 regression models m1-m18 with 80% prediction intervals (vertical lines). A horizontal dashed line at the average of the model forecasts is shown for reference only.

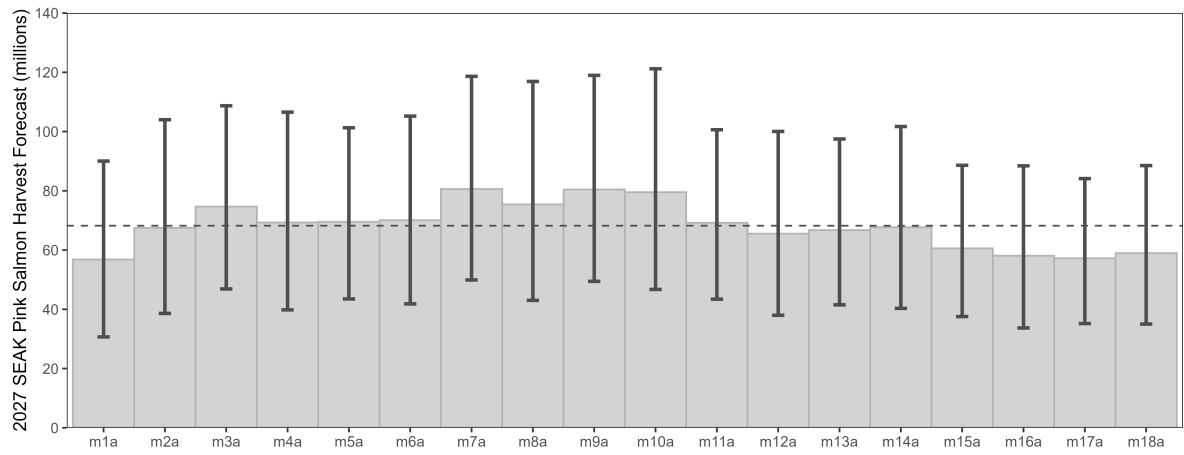


Figure 4: Bias-corrected forecasts (grey bars) for the 18 regression models m1a-m18a with 80% prediction intervals (vertical lines). A horizontal dashed line at the average of the model forecasts is shown for reference only.

Appendix C

Model Diagnostics

Model m13a included vessel-adjusted CPUE, an odd/even year factor, and satellite-derived SST data from northern Southeast Alaska in April, May, and June (NSEAK_SST_AMJ). Model diagnostics for model m13a included residual plots, a curvature (lack-of-fit) test, and influential observation diagnostics using Cook's distance (Cook 1977), the Bonferroni outlier test, and leverage (Table 7; Figure 5; Figure 6). Model diagnostics were used to identify observations that were potential outliers, had high leverage, or were influential (Zhang 2016).

Table 7: Detailed output for model m13a. Year is the harvest year. Fitted values (millions of fish) are bias-corrected.

Year	Harvest	Residuals	Hat values	Cooks distance	Std. residuals	Fitted values
1998	42.45	0.41	0.09	0.05	1.40	29.44
1999	77.82	-0.26	0.29	0.10	-0.99	105.62
2000	20.25	-0.24	0.11	0.02	-0.84	27.10
2001	67.02	0.07	0.09	0.00	0.24	65.56
2002	45.32	0.17	0.15	0.02	0.61	39.94
2003	52.47	-0.21	0.19	0.03	-0.76	68.09
2004	45.31	0.37	0.12	0.06	1.29	32.76
2005	59.12	0.07	0.10	0.00	0.23	57.98
2006	11.61	-0.36	0.13	0.05	-1.23	17.37
2007	44.80	0.04	0.09	0.00	0.12	45.37
2008	15.91	-0.31	0.11	0.03	-1.05	22.68
2009	38.02	-0.33	0.17	0.07	-1.17	55.41
2010	24.14	-0.18	0.09	0.01	-0.60	30.24
2011	58.88	-0.04	0.11	0.00	-0.12	64.02
2012	21.28	0.07	0.07	0.00	0.25	20.71
2013	94.72	0.36	0.11	0.05	1.22	69.66
2014	37.17	0.23	0.11	0.02	0.78	31.03
2015	35.09	-0.50	0.09	0.07	-1.68	60.40
2016	18.37	0.15	0.22	0.02	0.54	16.61
2017	34.73	-0.06	0.35	0.01	-0.26	38.87
2018	8.07	-0.38	0.14	0.07	-1.33	12.39
2019	21.14	-0.14	0.20	0.02	-0.50	25.47
2020	8.06	-0.34	0.18	0.08	-1.23	11.92
2021	48.53	0.51	0.12	0.11	1.77	30.42
2022	18.30	0.21	0.10	0.01	0.71	15.58
2023	48.45	0.65	0.17	0.27	2.30	26.58
2024	20.12	0.09	0.08	0.00	0.30	19.28

Table 7: Detailed output for model m13a. Year is the harvest year. Fitted values (millions of fish) are bias-corrected.

Year	Harvest	Residuals	Hat values	Cooks distance	Std. residuals	Fitted values
2025	21.37	-0.16	0.16	0.01	-0.56	26.24
2026	20.42	0.10	0.07	0.00	0.32	19.45

Cook's distance

Cook's distance is a measure of influence that combines leverage and the size of the residual,

$$D_i = \frac{e_{PSi}^2}{k+1} \cdot \frac{h_i}{1-h_i}, \quad (10)$$

where e_{PSi} is the standardized Pearson residual, h_i is the hat value (a measure of leverage), and k is the number of predictor variables in the model (Fox and Weisberg 2019). It measures the overall influence of the i th data point on all n fitted values; a large value indicates an influential observation. Cook's distance values greater than $4/(n-k-1)$, where n is the number of observations (29) and $k = 3$, were used as a benchmark for identifying potentially influential observations (Ren et al. 2016). Therefore, a cut-off of 0.16 was used (Figure 5 A).

Leverage

An observation that is distant from the average covariate pattern has high leverage (hat value). An observation with a leverage value h_i greater than 2 or 3 times p/n may be a concern (Ren et al. 2016), where p is the number of parameters in the model including the intercept (4) and n is the number of observations (29); here $p/n = 0.14$. Therefore, a leverage cut-off of $2p/n = 0.28$ was used; observations above this value may affect the model properties (e.g., summary statistics, standard errors, predicted values; Figure 5 B).

Significance of the temperature term

An F -test compared model m13a (full model) with a reduced model containing only the CPUE and odd/even year factor. The full model fit significantly better than the reduced model ($F = 17.3$, $p < 0.001$), indicating that the temperature term improves the model.

Influential observations, outliers, and lack of fit

Cook's distance was above the cut-off value for 1 year (Figure 5 A; 2023) and 2 observation(s) had high leverage values (Figure 5 B; 1999 and 2017). Based on the Bonferroni outlier test, none of the observations had a studentized residual with a significant Bonferroni-adjusted p -value (smallest adjusted $p = 0.53$, year 2023), suggesting that no single observation unduly affected the model fit; 2023 had the most extreme standardized residual (Figure 6 C; Table 7). Tukey's test for nonadditivity (Tukey 1949; Weisburg 2014) indicated some curvature in the residuals versus fitted values ($t = -2.25$, $p = 0.024$), consistent with the slightly curved smoother in Figure 6 D, suggesting some lack of fit.

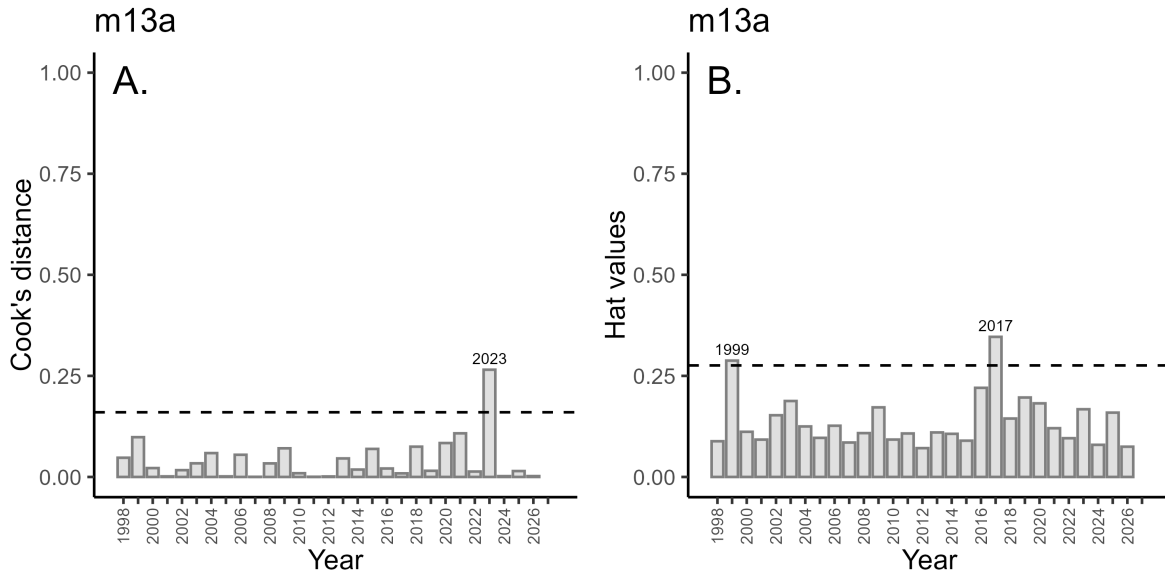


Figure 5: Diagnostic plots of influential observations from model m13a: A. Cook's distance (cut-off 0.16) and B. leverage values (cut-off 0.28). Year is the harvest year.

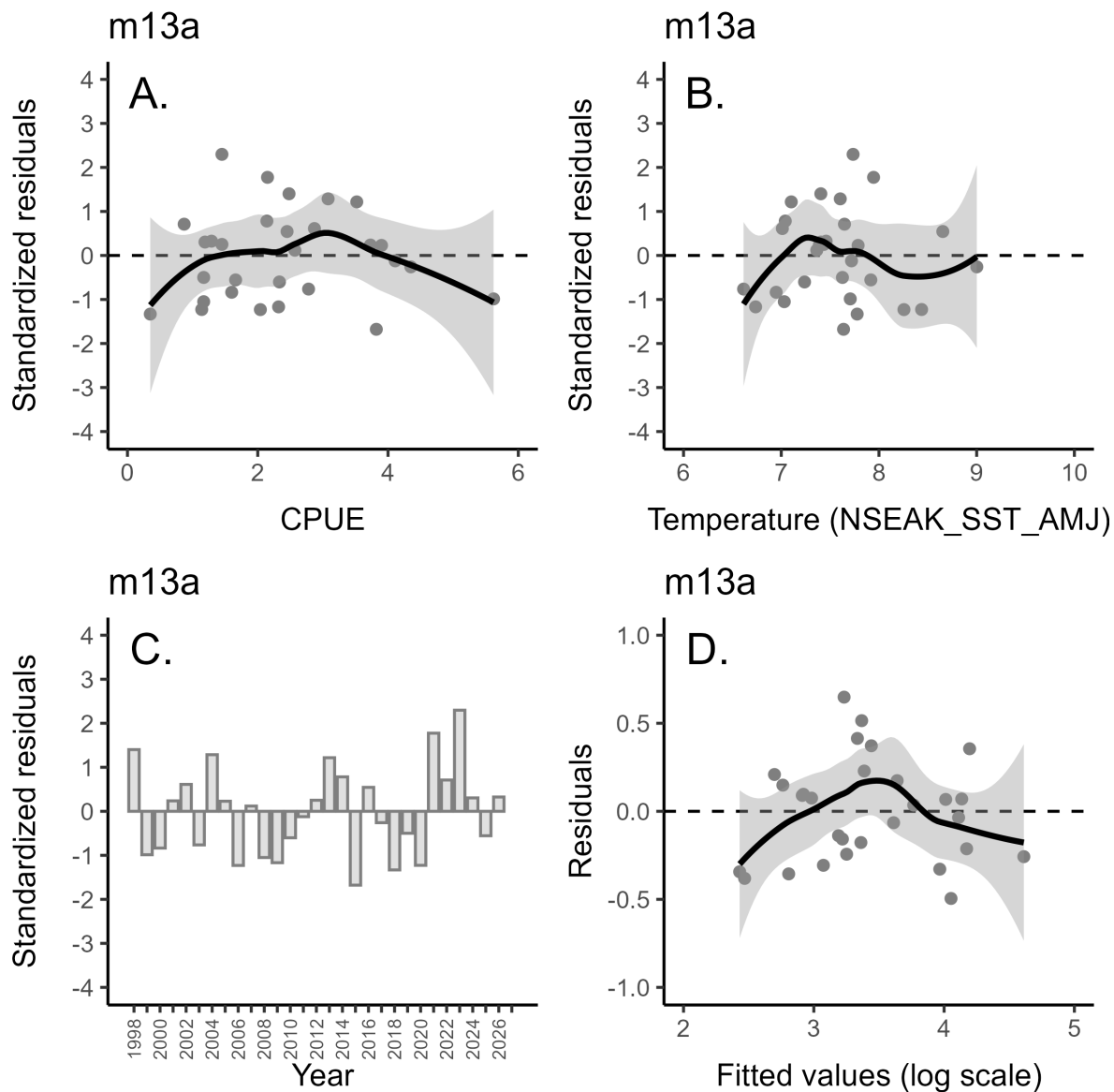


Figure 6: Standardized residuals for model m13a versus A. CPUE and B. temperature (NSEAK_SST_AMJ); C. standardized residuals by harvest year; and D. residuals versus fitted values (log scale). Positive residuals indicate that the observed harvest was larger than predicted by the model.